ASSIGNMENT : 4

NAME : GUALEEN
COURSE : B.TECH CS (DS)
ROLL NO. : 2301420051

1. Differentiate between SRAM and DRAM memory.

→ Static RAM (SRAM) and Dynamic RAM (DRAM) are two types of primary memory that differ in speed, cost and storage mechanism. SRAM stores data using flip-flops, while DRAM uses capacitors that require periodic refreshing.

* SRAM :
  - Uses flip-flops
  - It is very fast
  - Expensive
  - No refresh needed

* DRAM :
  - Uses capacitors
  - Slower than SRAM
  - Cheaper
  - Require refreshing

2. Locality of reference :

→ Locality of reference is the principle that programs tend to access a small portion of memory repeatedly rather than the entire memory randomly. This behavior helps improve performance in memory systems.

→ Types :

- Temporal Locality : Recently accessed data is likely to be used again.
- Spatial Locality : Nearby memory locations are likely to be accessed.

3. Cache Memory :

→ Cache Memory is a small, high-speed memory placed between the CPU and main memory to reduce access-time and improve system performance.

- It stores frequently used data/instructions.
- Faster than main memory.
- Reduces average memory access time.

4. Cache Indexing:

→ Cache indexing determines how data from main memory is mapped to cache locations.

- Direct Mapped Cache : one block maps to one cache line.
- Set Associative Cache : block can go to a set of locations.
- Fully Associative Cache : block can go anywhere in cache.

5. Cache Block Size:

→ Cache block size refers to the amount of data transferred between memory and cache at once.

* Larger block:
  - Exploits spatial locality.
  - Fewer compulsory misses.
* Smaller block:
  - Faster transfer
  - Less wasted data.

6. Cache Associativity:

→ Cache associativity refers to the number of possible locations in cache where a particular memory block can be placed. It defines the flexibility of mapping between main memory and cache.

→ Types:

- Direct mapped (1-way)
- Set associative (n-way)
- Fully associative

Improves cache hit rate.

7. Cache Size trade-off:

→ Cache size directly affects system performance by influencing how much data can be stored close to the CPU.

* Larger cache:
  Reduces miss rate
  Stores more frequently used data.
* Smaller cache:
  Faster access
  Lower cost

1. Memory Hierarchy Diagram

8. Writes policies:

→ Write policies determine how data is updated in cache and main memory during write operations.

- Write-Through:
  - Updates cache + main memory simultaneously
  - Simpler and more consistent.
  - Slower due frequent memory writes.
- Write-Back:
  - Updates only cache initially.
  - Main memory updated when block is replaced.
  - Faster but more complex.

9. Average Memory Access Time:

→ AMAT is the average time taken to access memory, considering both cache hits and misses.

- AMAT = Hit Time + (Miss Rate × Miss Penalty)

* Hit time: Time to access cache
* Miss Rate: Fraction of accesses that miss.
* Miss Penalty: Extra time to fetch from main memory.

10. Cache replacement (LRU):

→ Least Recently Used is a cache replacement policy that replaces the block that has not been used for the longest time.

- Based on temporal locality.
- Improves cache efficiency.
- Common in set-associative caches
- Slightly complex to implement.

11. Memory Interleaving:

→ Memory interleaving is a technique where memory is divided into multiple modules that can be ~~assced~~ accessed simultaneously to improve performance.

- Allows parallel memory access.
- Increases bandwidth.
- Reduces waiting time for CPU.

12. Magnetic Disk:

→ Magnetic Disks store data in a structured format using tracks and sector.

- Tracks:
  Circular path on the disk surface.
  Data is stored along these rings.
- Sectors:
  Subdivisions of tracks.
  Smallest unit of data storage.

13. Flash Memory:

→ Flash Memory is a type of non-volatile memory that retains data even when power is turned off. It is widely used in modern storage devices due to its speed and reliability. Faster than traditional magnetic disks and used in SSDs, USB drives, memory cards and based on electrical storage.

14. Memory - Mapped I/O:

→ Memory-mapped I/O is a technique where I/O devices share the same address space as main memory, allowing the CPU to access them like normal memory locations.

- No separate instructions needed.
- Uses load/store instructions.

15. I/O - Mapped I/O:

→ It uses a separate address space for I/O devices, different from main memory.

- Requires special I/O instructions.
- Does not use memory address space.
- Better isolation between memory & I/O.

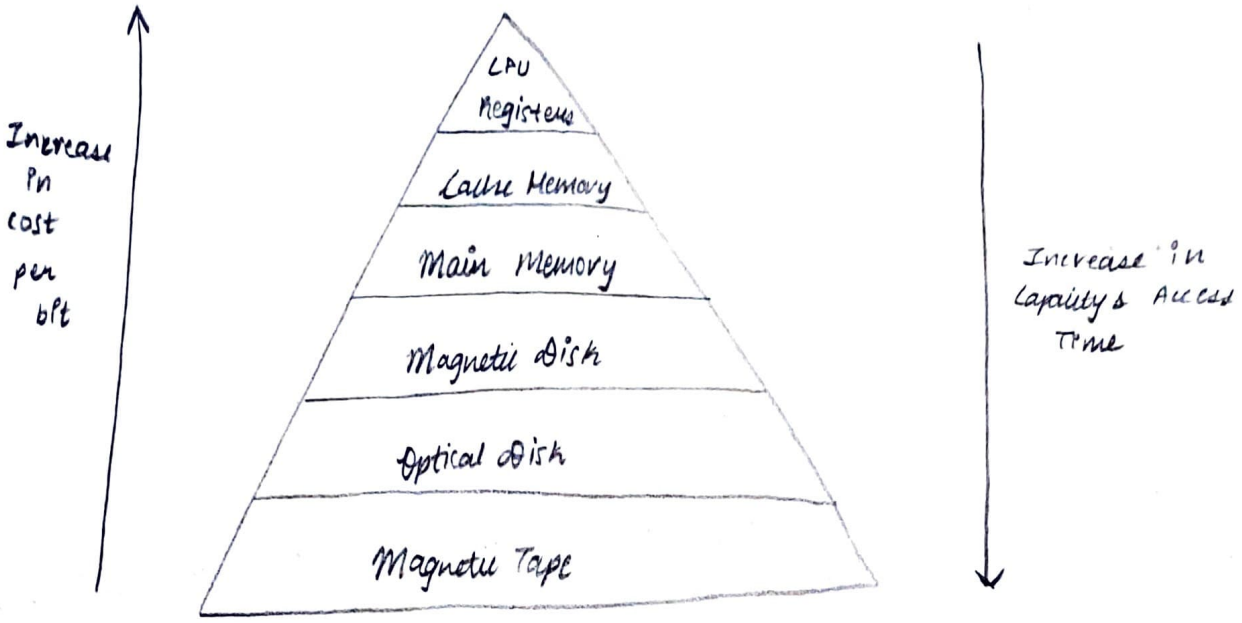
16. Programmed I/O:

→ Programmed I/O is a method where the CPU directly controls & waits for I/O operations, continuously checking device status.

- CPU actively involved in data transfer.
- Uses polling (busy waiting).
- Simple but inefficient.

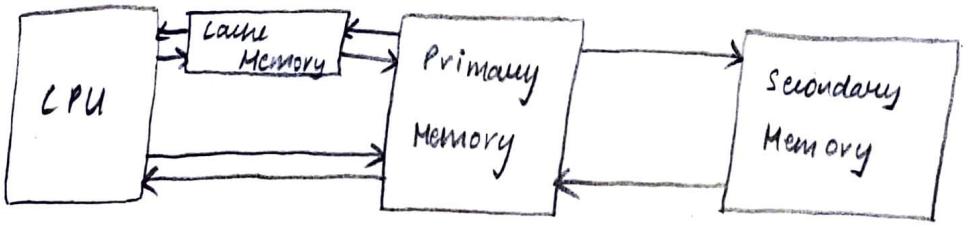
Memory Organization :

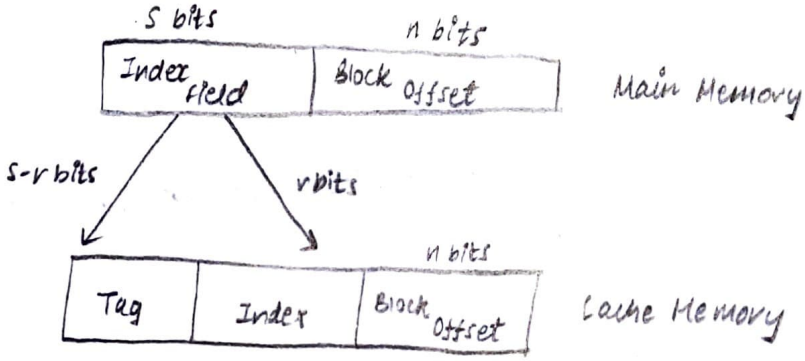
1. Memory Hierarchy Diagram



* Memory hierarchy organizes storage based on speed, cost, capacity. Faster memory is closer to CPU, while larger and slower memory is farther away.

2. Direct-Mapped Cache organisation:





* In direct-mapped cache, each memory block maps to exactly one cache line using the index field.

3. Write Through V/S Write Back:

→ Write Through:

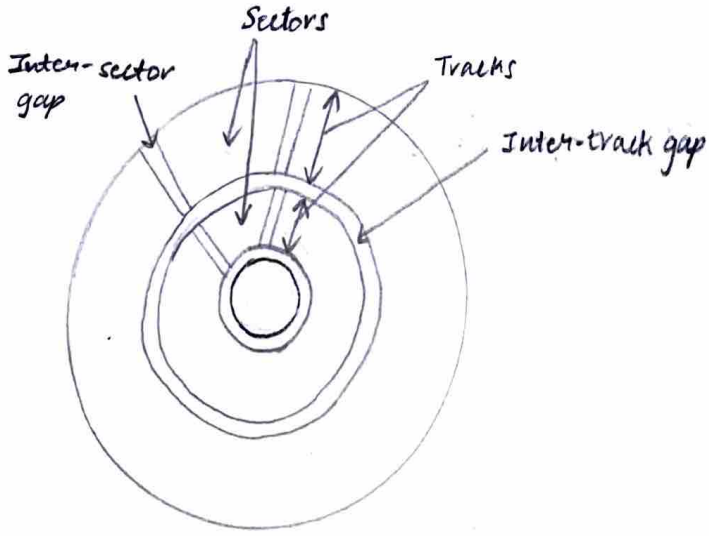
CPU → Cache → Main Memory
└→ Updated Immediately

→ Write Back:

CPU → Cache
└→ Main memory updated later on replacement

Write-through updates both cache and memory instantly, while write-back delays memory update until replacement.

4. Magnetic Disk Structure:



Magnetic disks store data in circular tracks, which are divided into smaller units called sectors.

- Track = circular path
- Sector = smallest storage unit
- Enables organized data access.

1. Repeated Access to Same Memory Location:

→ When a program repeatedly accesses the same memory locations, it follows the principle of Temporary (Temporal) locality. This means that recently accessed data is likely to be accessed again soon.

- Data reused within short time intervals.
- Basis for cache efficiency.
- Improves performance by keeping data in cache.

2. Larger Cache Block Size Slows Performance:

→ Although increasing cache block size, reduces miss rate, it can sometimes slow performance due to additional overhead.

- Higher Miss penalty.
- Cache pollution
- Reduces space for other useful data.

3. Write-Through Cache operation:

→ In a write-through cache operation, updates both the cache and the main memory simultaneously.

- Ensures data consistency.
- No stale data in memory.
- Slower due to frequent memory writes.
- Often uses write buffer to reduce delay.

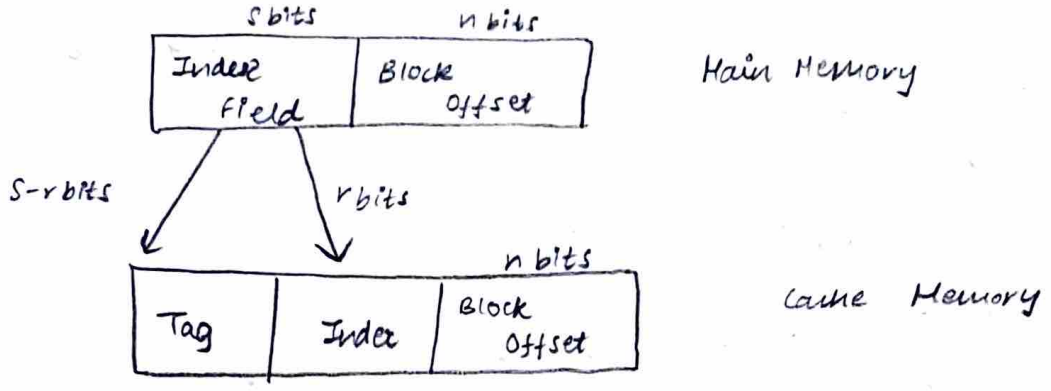
4. High-Speed Data Transfer without CPU:

→ When a device like a network card transfers large data blocks directly to memory without CPU involvement, the technique used is Direct Memory Access.

- Data Transfer handled by DMA controller.
- CPU only initiates transfer
- Improves system performance.

Section - D

1. Cache Memory Organisation:



→ The index selects the cache line, the tag is compared to check if it's the correct block (hit/miss) and the offset selects the exact data within the block.

2. AMAT :

→ Average Memory Access Time = Hit Time + (Miss Rate × Miss Penalty)

where,

- Hit time : Time to access cache (should be minimal)
- Miss Rate : Frequency of cache misses.
- Miss Penalty : Time to fetch data from lower memory

Even a small increase in miss rate can drastically increase AMAT, making cache design critical.

3. I/O Data Transfer Techniques :

- Programed I/O :

CPU ⟶ I/O Device

(CPU continuously checks and waits)

- Interrupt I/O driven:

CPU ⟶ I/O signal

CPU ⟵ Interruptt Signal

CPU works, device interrupts when ready

- Direct Memory Access :

I/O Device ⇄ Memory

↓

CPU (only initiates transfer)

Direct memory Access system performance by freeing the CPU and enabling high-speed data transfer.